

Guoheng Sun

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Research interests

My primary research focus is on **improving the efficiency of LLMs/VLMs/VLAs**. In addition, I am interested in **trustworthy AI** (e.g., transparency in AI service billing), **VLA models** (improving spatial reasoning, design efficient architectures), and **post-training techniques for LLMs** (e.g., enhancing reasoning ability, multi-turn interaction, and tool calling).

Education

University of Maryland, College Park

PH.D. STUDENT IN COMPUTER ENGINEERING

8/2024 - 5/2028 (expected)

Advisor: [Prof. Ang Li](#)

Sichuan University

SCHOOL OF CYBER SCIENCE AND ENGINEERING

9/2020 - 6/2024

Selected Publications

*Equal Contribution †Equal Supervision

[VLA / Efficiency] G. Sun, T. Du, K. Feng, C. Luo, X. Ding, Z. Shen, Z. Wang, Y. He, A. Li, “ROCKET: Residual-Oriented Multi-Layer Alignment for Spatially-Aware Vision-Language-Action Models”. arXiv preprint arXiv:2602.17951, 2026. [\[Link\]](#)

[VLA/Efficiency] G. Sun, K. Feng, S. He, X. Gong, Y. He, Z. Wang, Z. Shen, W. Ye, R. R. Kompella, G. Liu, A. Li, “Drop-Then-Recovery: How Redundant Are Vision-Language-Action Models?”. arXiv preprint arXiv:2606.27755, 2026. [\[Link\]](#)

[VLA] Z. Wang, B. Wang, H. Zhang, T. Du, T. Chen, G. Sun, Y. He, Z. Shen, W. Ye, A. Li, “Vision-Language-Action in Robotics: A Survey of Datasets, Benchmarks, and Data Engines”. Transactions on Machine Learning Research (TMLR), 2026. [\[Link\]](#)

[Efficiency] S. He, G. Sun, H. Zhang, Y. Fu, A. Li, “Demystifying When Pruning Works via Representation Hierarchies”. International Conference on Machine Learning (ICML), 2026. [\[Link\]](#)

[Security] G. Sun*, Y. Fu*, H. Yang, J. Huang, R. Zhang, H. Wang, “Enhancing the Security of Large Character Set CAPTCHAs Using Transferable Adversarial Examples”. IEEE Transactions on Dependable and Secure Computing (TDSC), 2025. [\[Link\]](#)

[Reasoning LLM Audit] G. Sun, Z. Wang, B. Tian, M. Liu, Z. Shen, S. He, Y. He, W. Ye, Y. Wang, A. Li, “CoIn: Counting the Invisible Reasoning Tokens in Commercial Opaque LLM APIs”. Findings of the Association for Computational Linguistics: EMNLP, 2026. [\[Link\]](#)

[Reasoning LLM Audit] G. Sun*, Z. Wang*, X. Zhao, B. Tian, Z. Shen, Y. He, J. Xing, A. Li, “Position: Invisible Tokens, Visible Bills: The Urgent Need to Audit Hidden Operations in Opaque LLM Services”. International Conference on Machine Learning (ICML), 2026. [\[Link\]](#)

[Reasoning LLM Audit / RL] Z. Wang*, G. Sun*, Y. He, Z. Shen, B. Tian, A. Li, “Predictive Auditing of Hidden Tokens in LLM APIs via Reasoning Length Estimation”. arXiv preprint arXiv:2508.00912, 2025. [\[Link\]](#)

[Efficiency] S. He*, G. Sun*, Z. Shen, A. Li, “Uncovering the Redundancy in Transformers via a Unified Study of Layer Dropping”. Transactions on Machine Learning Research (TMLR), 2026. [\[Link\]](#) [\[Code\]](#)

[Reasoning / Code / RL] Y. Wang*, G. Sun*, W. Ye, G. Qu, A. Li, “VeriReason: Reinforcement Learning with Testbench Feedback for Reasoning-Enhanced Verilog Generation”. Great Lakes Symposium on VLSI (GLSVLSI), 2026. [\[Link\]](#)

[Reasoning / Code / RL] Z. Yao*, G. Sun*, L. Borchmann, Z. Shen, M. Deng, B. Zhai, H. Zhang, A. Li, Y. He, “Arctic-Text2SQL-R1: Simple Rewards, Strong Reasoning in Text-to-SQL”. Findings of the Association for Computational Linguistics: ACL, 2026. [\[Link\]](#) [\[Code\]](#)

[Reasoning / Code / RL] Z. Song, Y. Liu, C. Chen, G. Sun, J. Yin, C. Ho, A. Li, H. Ren, C. Yu, “TOPCELL: Topology Optimization of Standard Cell via LLMs”. 63rd ACM/IEEE Design Automation Conference (DAC), 2026. [\[Link\]](#)

[Reasoning / Code / RL] Y. Wang, W. Ye, P. Guo, Y. He, Z. Wang, B. Tian, S. He, G. Sun, Z. Shen, S. Chen, A. Srivastava, Q. Zhang, G. Qu, A. Li, “SymRTL0: Enhancing RTL Code Optimization with LLMs and Neuron-Inspired Symbolic Reasoning”. NeurIPS 2025. [\[Link\]](#)

[Efficiency] X. Zhao*, G. Sun*, R. Cai*, Y. Zhou*, P. Li*, P. Wang, B. Tan, Y. He, L. Chen, Y. Liang, B. Chen, B. Yuan, H. Wang†, A. Li†, Z. Wang†, T. Chen†, “Model-GLUE: Democratized LLM Scaling for A Large Model Zoo in the Wild”. NeurIPS 2024 Datasets and Benchmarks [\[Link\]](#)

[Efficiency] Z. Shen, Y. He, Z. Wang, Y. Zhang, G. Sun, W. Ye, A. Li, “EdgeLoRA: An Efficient Multi-Tenant LLM Serving System on Edge Devices”. 23rd ACM International Conference on Mobile Systems, Applications, and Services (MobiSys), 2025. [\[Link\]](#)

[Efficiency] S He, T Ge, G Sun, et al. “Router-Tuning: A Simple and Effective Approach for Enabling Dynamic-Depth in Transformers”. EMNLP (Main) 2025. [\[Link\]](#)

[Security] Z Wang, Z Shen, Y He, G Sun, et al. “Flora: Federated fine-tuning large language models with heterogeneous low-rank adaptations”. NeurIPS 2024. [\[Link\]](#)

[Efficiency] Y He, Z Wang, Z Shen, G Sun, et al. “SHED: Shapley-Based Automated Dataset Refinement for Instruction Fine-Tuning”. NeurIPS 2024. [\[Link\]](#)

Research Experience

Research Internship at Sony AI

Mentor: Lingjuan Lyu

7 / 2026 – 10 / 2026

- Streaming inference for World-Action Models (WAMs), which condition action generation on predicted future observations and are therefore expensive to run in closed loop.
- Proposed STAIRCASE POLICY, which splits a large action chunk into sub-chunks held at staggered denoising stages: near-term actions are executed as soon as they are ready while later ones keep being refined, and at each sub-chunk boundary the future latent is re-predicted from the newest observation to update all unexecuted actions.
- The resulting S-WAM reaches 97.7% average success on LIBERO at 369.1 executed actions per second, a 4.6× throughput gain over conventional execution, and 87.8% on LIBERO-Plus.

Research Internship at Snowflake AI Research

Mentor: Zhewei Yao & Yuxiong He

5 / 2025 – 8 / 2025

- Verl, Ray etc.
- Using reinforcement learning to train a SOTA text-to-SQL model. Arctic-text2sql-r1-32B achieved Top 1 on the BIRD-SQL Leaderboard (as of June 2025).
- Enhancing LLM text-to-SQL capability through reinforcement learning by introducing tool calls in multi-turn dialogues.
- Ongoing project: Building a multi-turn dialogue and tool-calling Text2SQL model with online knowledge injection, enabling the model to execute tools during multi-turn conversations and to learn knowledge it has never been trained on before.

Research Internship at the LLM360 Team/MBZUAI

Mentor: Hongyi Wang

9 / 2024 – 12 / 2024

- Megatron, TorchTitan, etc.
- Reproduce the scaling law of LLaMA 3 and explore scaling laws under overtraining scenarios. Contribute to model size selection, FLOPs calculation, fitting IsoFLOPs curves, and predicting model loss on test sets as well as performance on downstream tasks.
- Process mathematical data, including deduplication, format modification, and fixing formulas and tables.
- The outcomes have been consolidated and presented in [\[Slides\]](#).

Research Internship at UMD (CASE Lab)

Advisor: Ang Li

6 / 2023 – 6 / 2024

- Studied architectural redundancy in transformers: the more computationally intensive attention layers are the most redundant, especially in the middle of the network, and dropping them causes little degradation. For 70B LLMs, removing about half of the attention layers gives roughly 50% speedup while retaining over 95% of the performance.
- Benchmarked existing model merging and model mixture methods, and combined the two so that models with different architectures, initializations, and hard-to-merge weights can be reused.
- The resulting Model-GLUE method improves average performance by 5.61% over the best single model in the model zoo.

Skills

- Languages** Python, C#, C++, C, JAVA, JavaScript, etc.
- Tool Kits** Git, Bash, Docker, MySQL, etc.
- Others** PyTorch, TensorFlow, $\LaTeX$, Unity, Sklearn, Blender, etc.

Awards

- 2025 **Qualcomm Innovation Fellowship** [\[Link\]](#), Qualcomm
- 2024 **The Dean's Fellowship**, University of Maryland, College Park
- 2022 **National Scholarship**, Ministry of Education (the highest honor scholarship in China)